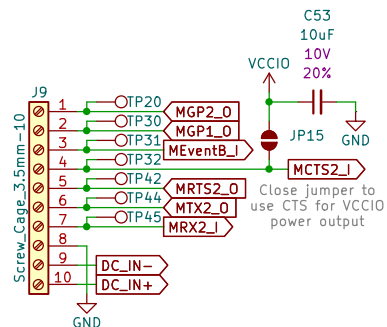
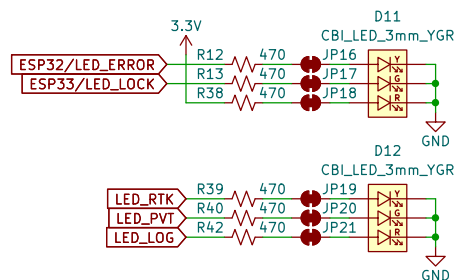


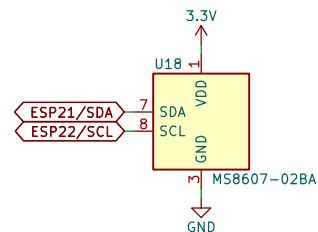
I/O Connector



LEDs

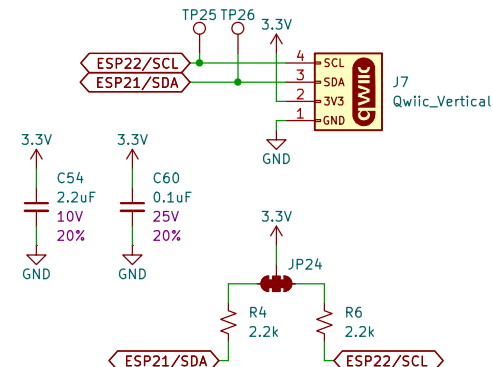


PHT Sensor – MS8607-02BA



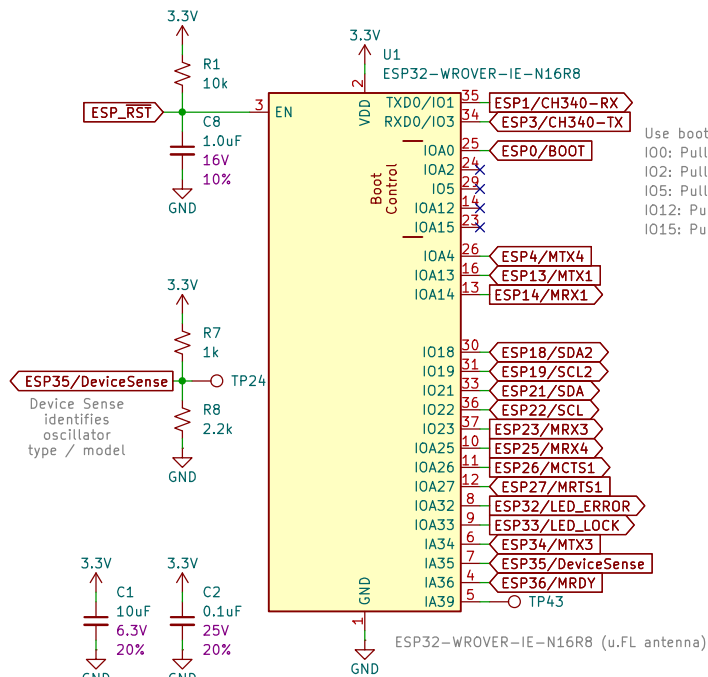
MS8607 I²C addresses:
P&T: 0x76 (unshifted)
H: 0x40 (unshifted)

Qwiic I²C (for OLED)



SSD1306 default I²C address: 0x3D (unshifted)

ESP32-WROVER



Use boot control pins with caution: 0, 2, 5, 12, 15
IO0: Pull-up at boot. Can be used a stat LED.
IO2: Pull-down at boot. Boot mode.
IO5: Pull-up at boot. SDIO timing.
IO12: Pull-down at boot. LDO voltage.
IO15: Pull-up. TX0 debug active.

Power File: Power.kicad_sch
PowerPath File: PowerPath.kicad_sch
USB File: USB.kicad_sch
GNSS File: GNSS.kicad_sch
Ethernet File: Ethernet.kicad_sch
LevelShifting File: LevelShifting.kicad_sch
LevelShifting_10MHz File: LevelShifting_10MHz.kicad_sch
Oscillator File: Oscillator.kicad_sch



SPARKPNT

Designed by: P.C.
SparkFun Electronics

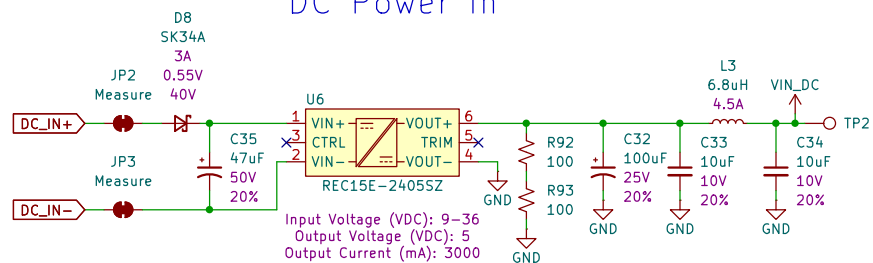
Sheet: /
File: SparkPNT_GNSSDO_Plus.kicad_sch

Title: GNSSDO Plus (mosaic-T, STP3593LF)

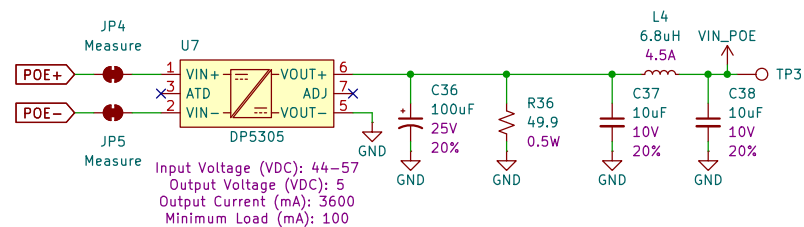
Size: USLetter Date: 2025-06-06
KiCad E.D.A. 9.0.1

Rev: v02
Id: 1/9

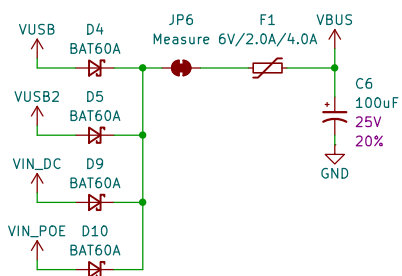
DC Power In



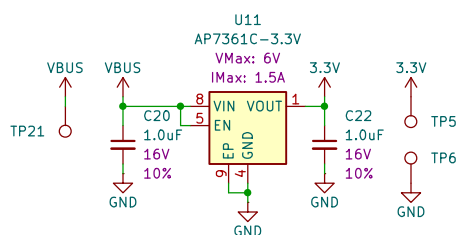
Power Over Ethernet



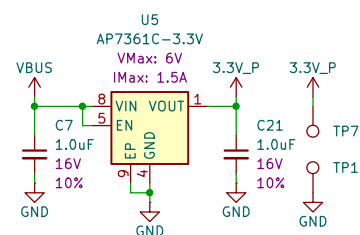
Power Mux



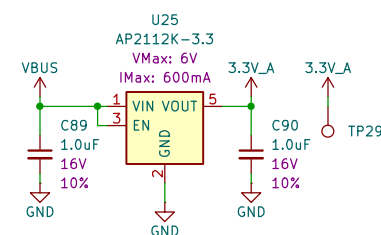
Main 3.3V



Peripheral 3.3V



Analog 3.3V



Sheet: /Power/
File: Power.kicad_sch

Title: GNSSDO Plus - Power

Size: USLetter Date:

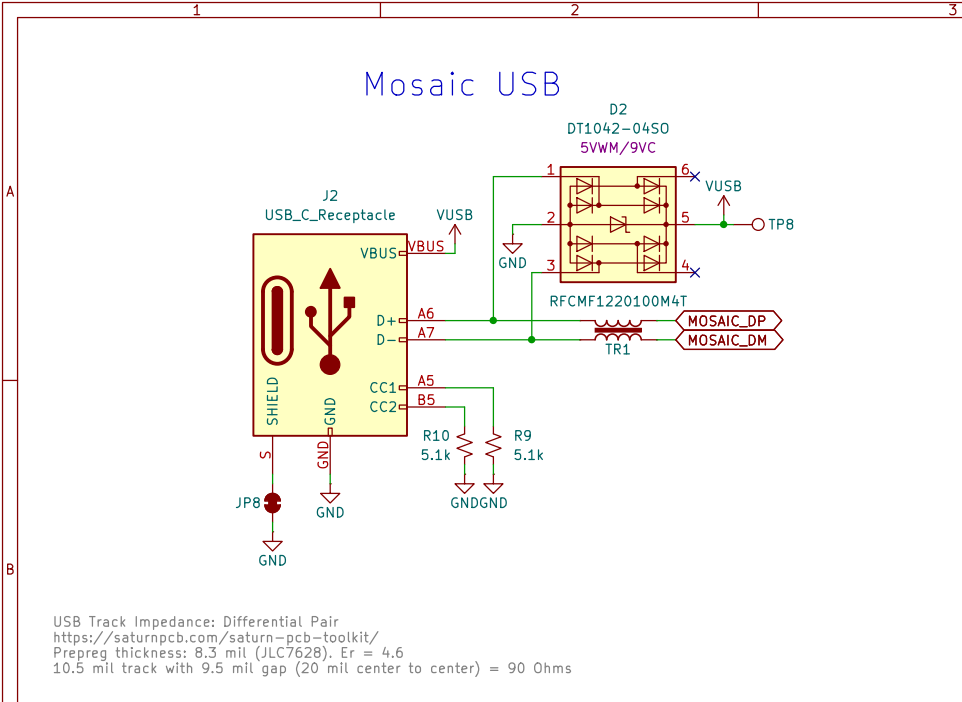
KiCad E.D.A. 9.0.1

Rev:

Id: 2/9

Mosaic USB

USB Track Impedance: Differential Pair
<https://saturnpcb.com/saturn-pcb-toolkit/>
Prepreg thickness: 8.3 mil (JLC7628). Er = 4.6
10.5 mil track with 9.5 mil gap (20 mil center to center) = 90 Ohms

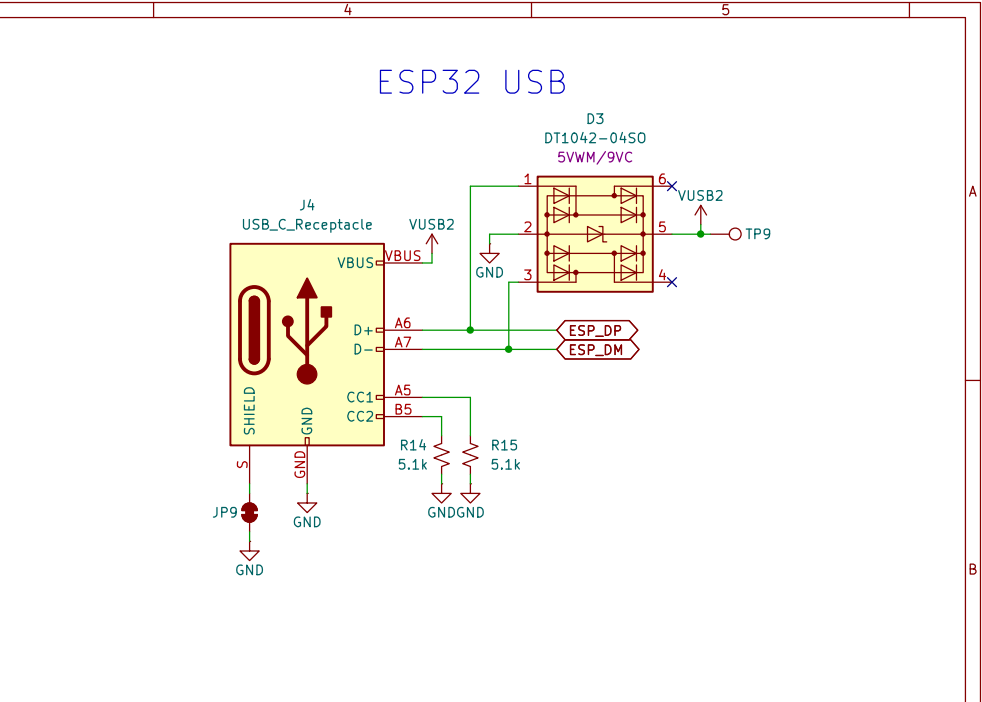


Mosaic USB

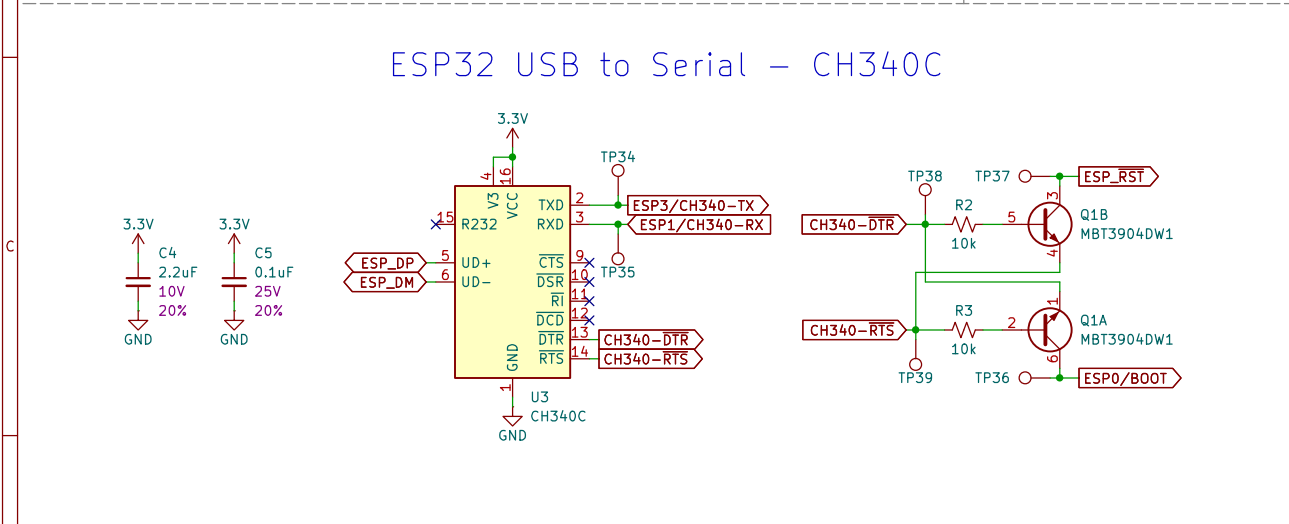
USB Track Impedance: Differential Pair
<https://saturnpcb.com/saturn-pcb-toolkit/>
Prepreg thickness: 8.3 mil (JLC7628). Er = 4.6
10.5 mil track with 9.5 mil gap (20 mil center to center) = 90 Ohms

ESP32 USB

The diagram illustrates the USB interface for an ESP32 module. A USB-C receptacle (J4) is connected to the module's pins. The receptacle's VBUS pin is connected to the module's VBUS pin. The receptacle's D+ and D- pins are connected to the module's A6 and A7 pins, respectively. The receptacle's CC1 and CC2 pins are connected to the module's A5 and B5 pins, which are then connected to ground through 5.1k resistors (R14 and R15). The receptacle's shield pin is connected to ground through a 5.1k resistor (R14). The module's VBUS pin is connected to a USB2 port (VUSB2) through a 5VWM/9VC diode (D3). The module's A6 and A7 pins are connected to the ESP32 module's ESP_DP and ESP_DM pins, respectively. The module's A5 and B5 pins are connected to ground through 5.1k resistors (R14 and R15). The module's shield pin is connected to ground through a 5.1k resistor (R14). The module's VBUS pin is connected to a USB2 port (VUSB2) through a 5VWM/9VC diode (D3). The module's A6 and A7 pins are connected to the ESP32 module's ESP_DP and ESP_DM pins, respectively. The module's A5 and B5 pins are connected to ground through 5.1k resistors (R14 and R15). The module's shield pin is connected to ground through a 5.1k resistor (R14).



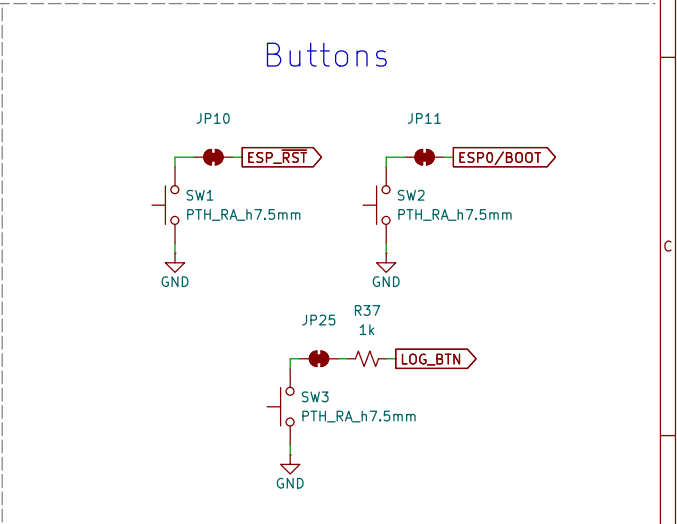
The diagram illustrates the internal circuit of an ESP32 USB to Serial module. It features a central CH340C IC (U3) which acts as a USB-to-UART bridge. The IC is powered by a 3.3V supply (V3) and has its ground (GND) connected to the module's common ground. The TXD and RXD pins of the CH340C are connected to the TX and RX pins of the ESP32, respectively. The CH340C also has pins for DTR and RTS, which are connected to the DTR and RTS pins of the ESP32. The module includes two 10V, 20% tolerance capacitors (C4 and C5) for decoupling the 3.3V supply. The ESP32 is shown with its pins for ESP_DP, ESP_DM, and ESP_RST. The CH340C is also connected to the module's ground through a 10k resistor (R2) and a 10k resistor (R3). The module is labeled with various test points (TP34, TP35, TP37, TP38, TP39, TP36) and component values (R2, R3, C4, C5).



Buttons

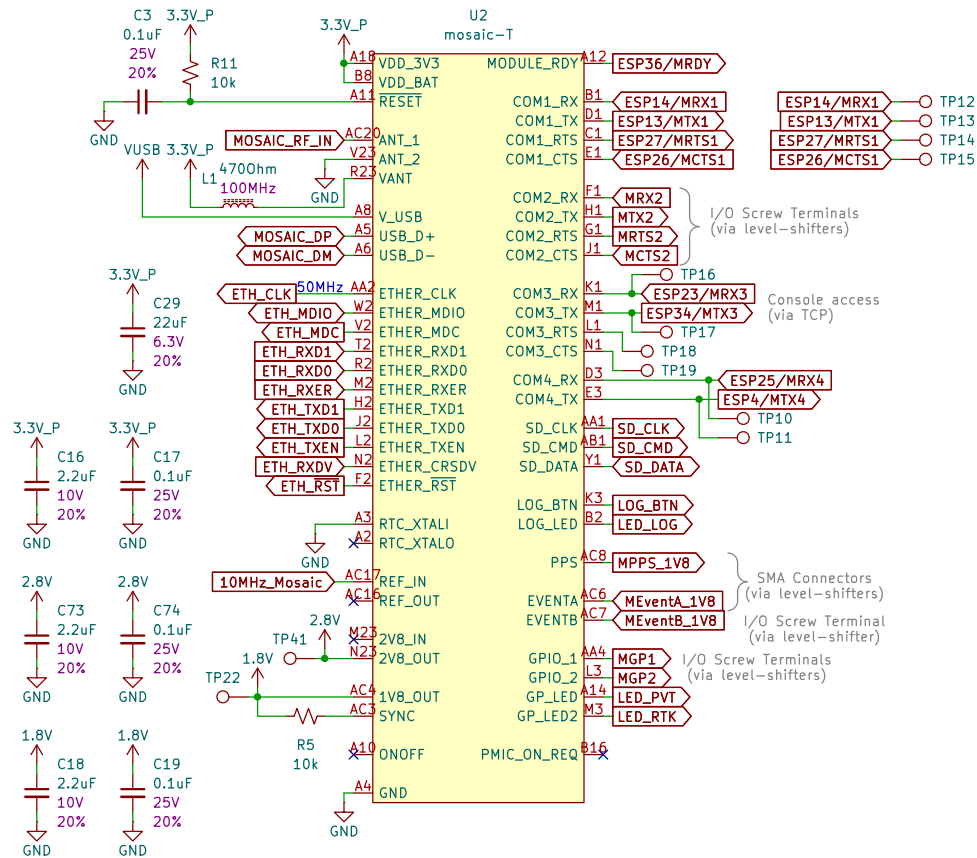
The image displays three separate circuit diagrams for button connections, each featuring a green wire, a red button symbol, and a red arrow pointing to a specific pin or component.

- Top Left Diagram:** A green wire connects a red button symbol to a red arrow pointing to a pin labeled **ESP_RST**. The button is connected to a switch labeled **SW1** with the text **PTH_RA_h7.5mm** below it. The other terminal of the switch is connected to a ground symbol labeled **GND**.
- Top Right Diagram:** A green wire connects a red button symbol to a red arrow pointing to a pin labeled **ESP0/BOOT**. The button is connected to a switch labeled **SW2** with the text **PTH_RA_h7.5mm** below it. The other terminal of the switch is connected to a ground symbol labeled **GND**.
- Bottom Diagram:** A green wire connects a red button symbol to a red arrow pointing to a component labeled **LOG_BTN**. The button is connected to a switch labeled **SW3** with the text **PTH_RA_h7.5mm** below it. The other terminal of the switch is connected to a ground symbol labeled **GND**. Above the switch, a resistor labeled **R37** with a value of **1k** is connected to a pin labeled **JP25**.

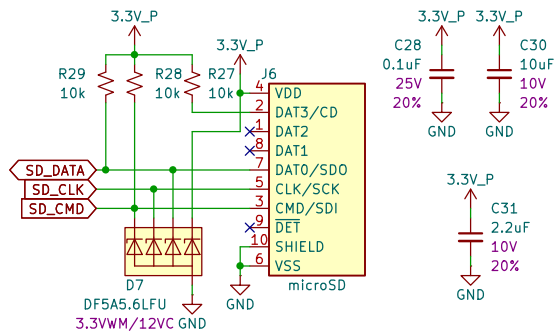


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Id: 3/9

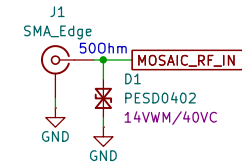
mosaic Tri-band GNSS



microSD



GNSS Antenna



Microstrip Calculation:
 Copper Thickness (1oz): 1.4mil/0.035mm
 Board thickness: 1.6mm
 Dielectric thickness (layer 1 to 2): 0.2mm
 Er: 4.6
 Polygon Isolation: 6mil/0.1524mm
 RF Trace Width: 13mil/0.33mm
<https://chemandy.com/calculators/coplanar-waveguide-with-ground-calculator.htm>

Sheet: /GNSS/
 File: GNSS.kicad_sch

Title: GNSSDO Plus - GNSS

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Ethernet

J5
MagJack_PoE

3.3V_P

JP12 R17 1k SPEED

JP13 R18 1k LINK

TD+ TD- RD+ RD-

C10 0.1uF 25V 20%

C9 1nF 2kV 10%

GND-ISO

GND

D6

DT1042-0450 5VWM/9V

R21 49.9

C12 0.1uF 25V 20%

C11 0.1uF 25V 20%

R19 49.9

R20 49.9

VDDA

C14 0.1uF 25V 20%

C13 2.2uF 10V 20%

R24 10k

ETH_RST

TP23

L2 4700hm 100MHz

U4
KSZ8041NL/I

VDDIO_3.3 VDDA_3.3 VDDP_1.8

RST

TX+ TX- RX+ RX-

RD+ RD- TX_EN TXD0 TXD1 TXD2 TXD3 TXC

CRS/CONFIG1 COL/CONFIG0 LED1/SPEED LED0/NWAYEN

GND

R35 4.7k R34 4.7k R23 10k

SPEED LINK

3.3V_P

R26 13k R25 13k

C15 100pF 50V 10%

ETH_CLK 50MHz

ETH_MDIO ETH_MDC ETH_RXD1 ETH_RXD0 ETH_RXDV ETH_RXER ETH_TXEN ETH_TXD0 ETH_TXD1

C26 0.1uF 25V 20%

C27 10uF 10V 20%

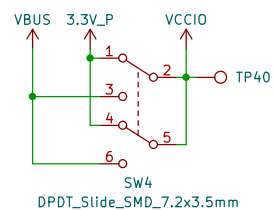
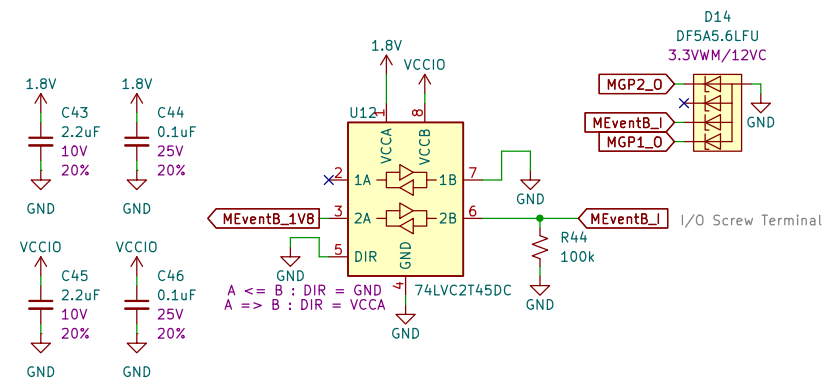
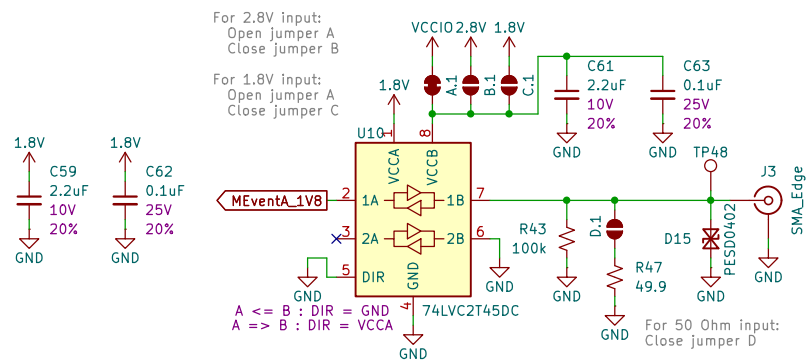
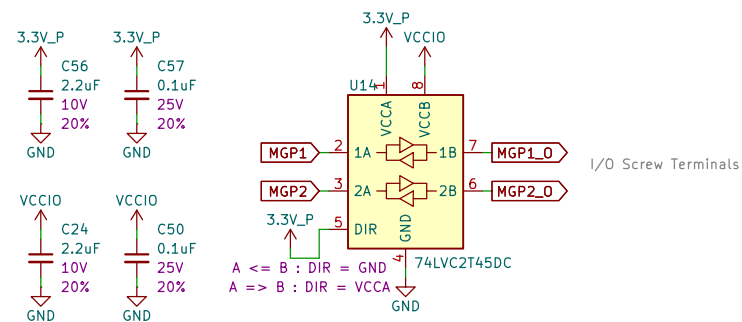
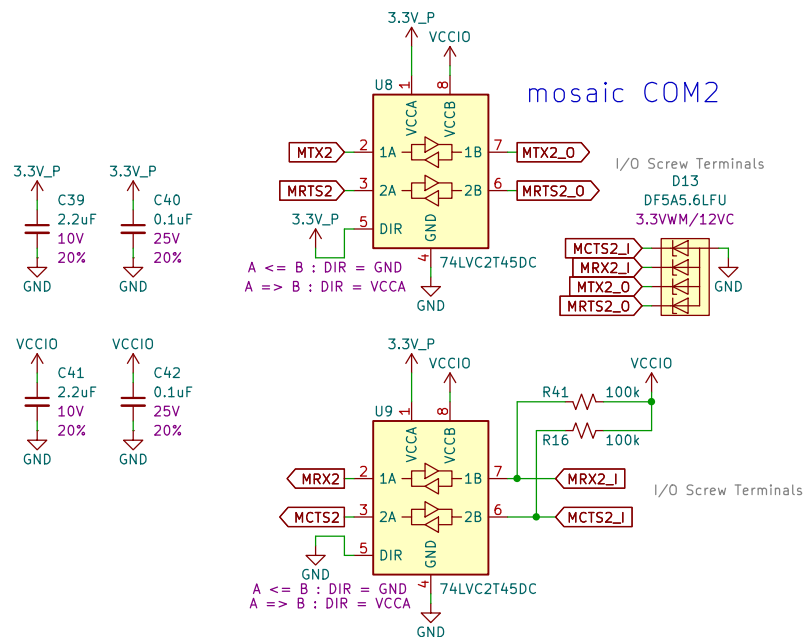
C23 0.1uF 25V 20%

C25 10uF 10V 20%

VDDA VDDA 3.3V_P 3.3V_P

Sheet: /Ethernet/	
File: Ethernet.kicad_sch	
Title: GNSSDO Plus - Ethernet	
Size: USLetter	Date:
KiCad E.D.A. 9.0.1	Rev:
Id: 5/9	

Sheet: /Ethernet/ File: Ethernet.kicad_sch	
Title: GNSSDO Plus – Ethernet	
Size: USLetter	Rev:
Date:	
KiCad E.D.A. 9.0.1	Id: 5/9

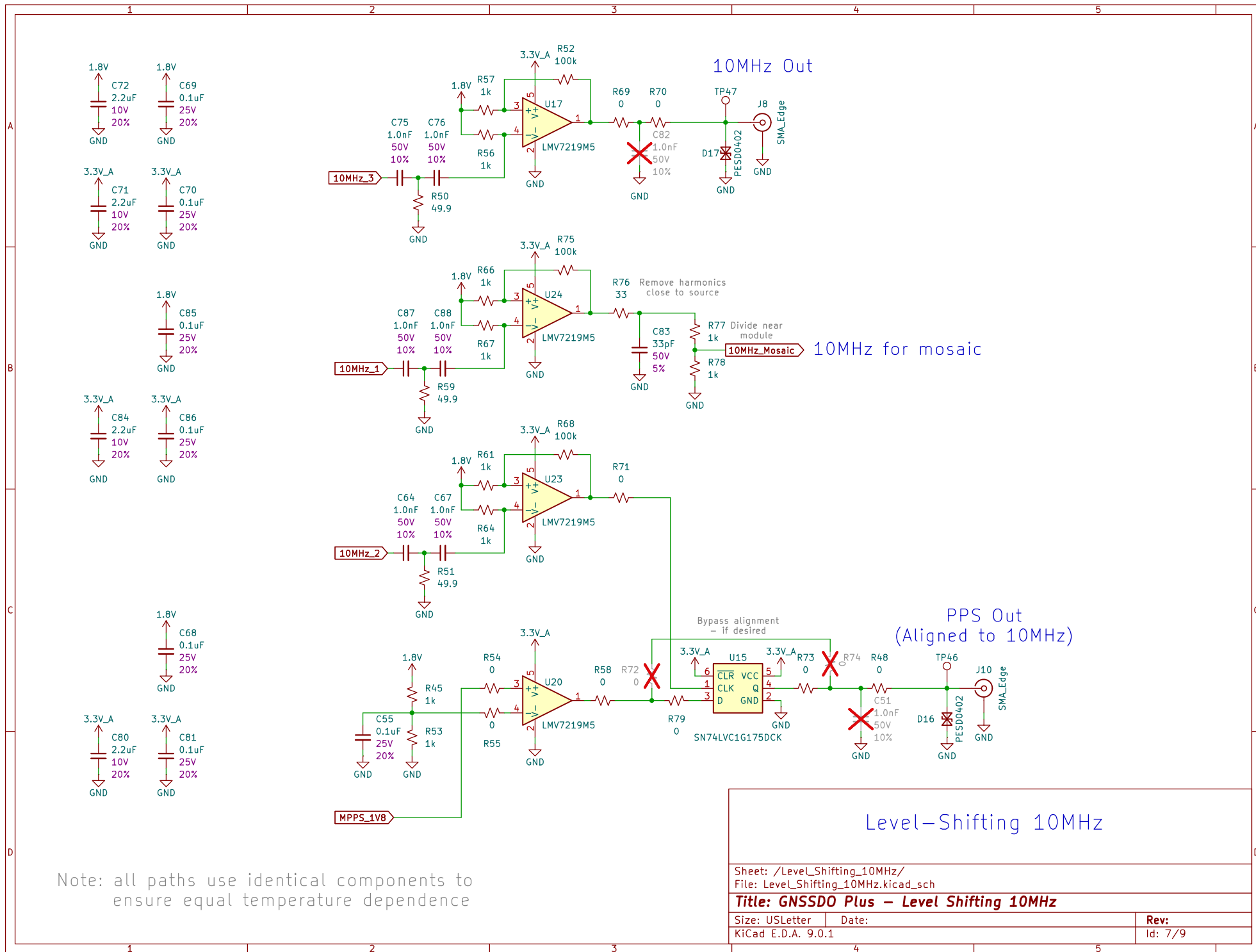


Sheet: /Level_Shifting/
File: Level_Shifting.kicad_sch

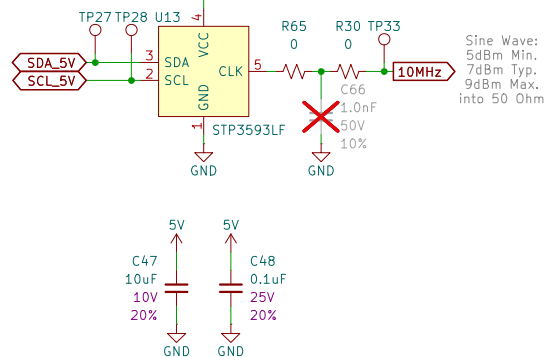
Title: GNSSDO Plus – Level Shifting

Size: USLetter	Date:
KiCad E.D.A. 9.0.1	

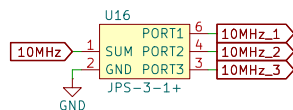
Rev:
Id: 6/9



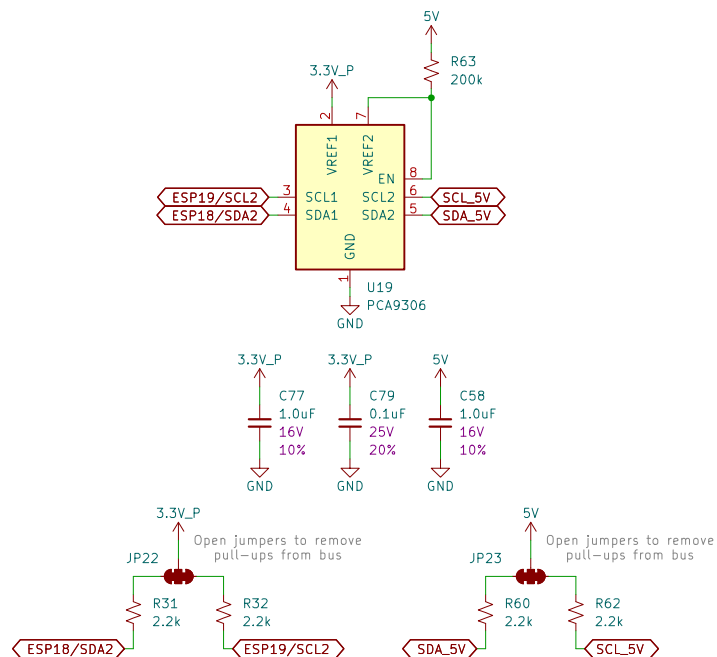
Supply Voltage: 5.0V (4.75V Min., 5.25V Max.)
Current Consumption: 1500mA (Warm Up), 600mA (Steady State)



Typical Total Loss: 5.0dB at 10MHz



I²C Level Shifting – PCA9306

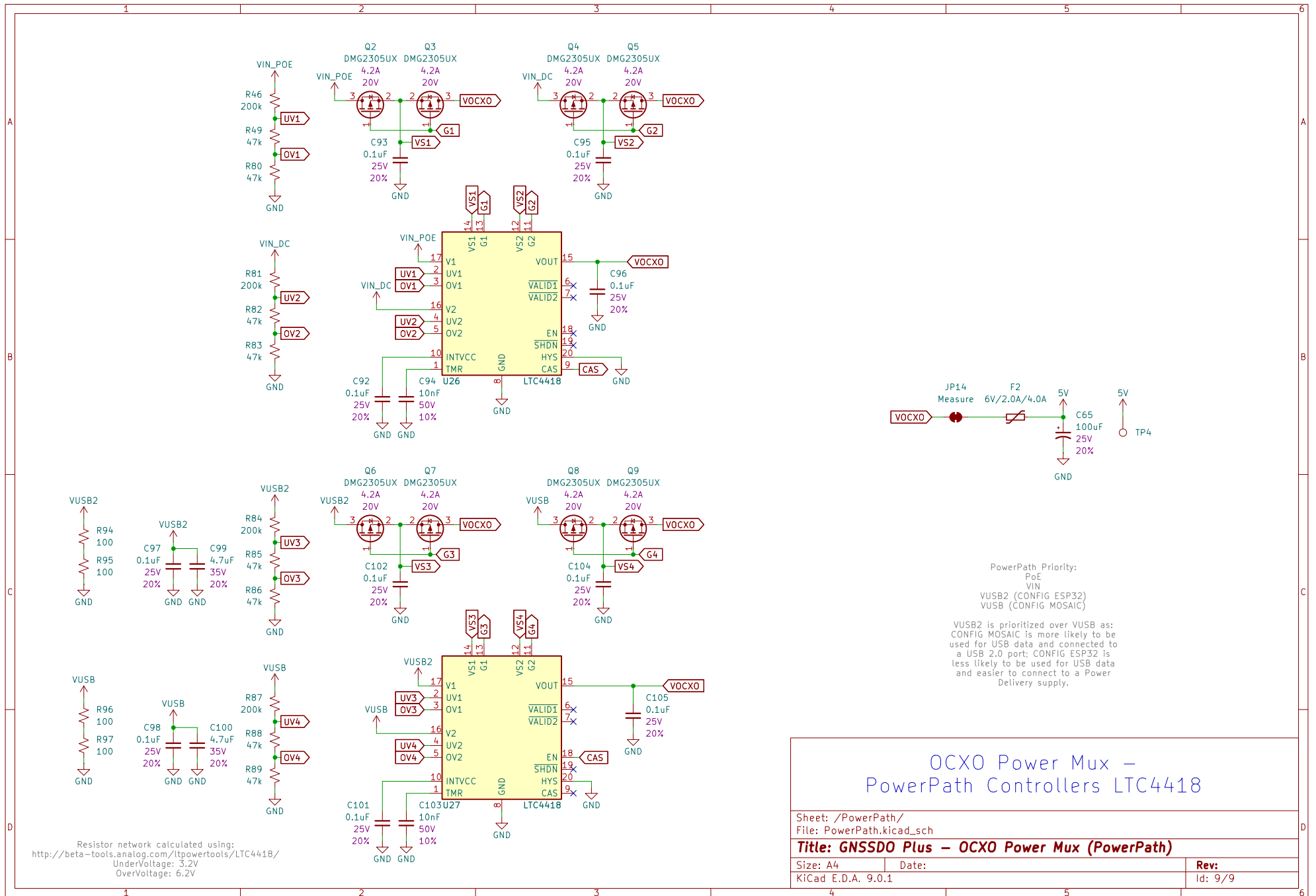


Title: GNSSDO Plus – Oscillator

Date:

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OXO Power Mux – PowerPath Controllers LTC4418

Sheet: /PowerPath/
 File: PowerPath.kicad_sch

Title: GNSSDO Plus – OXO Power Mux (PowerPath)

Size: A4

Date:

Rev:

KiCad E.D.A. 9.0.1

Id: 9/9